

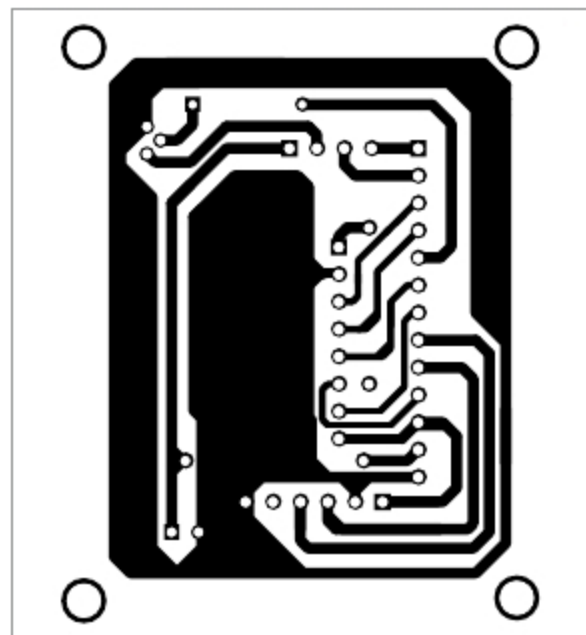


Fig. 7: Actual-size PCB layout of air quality monitor

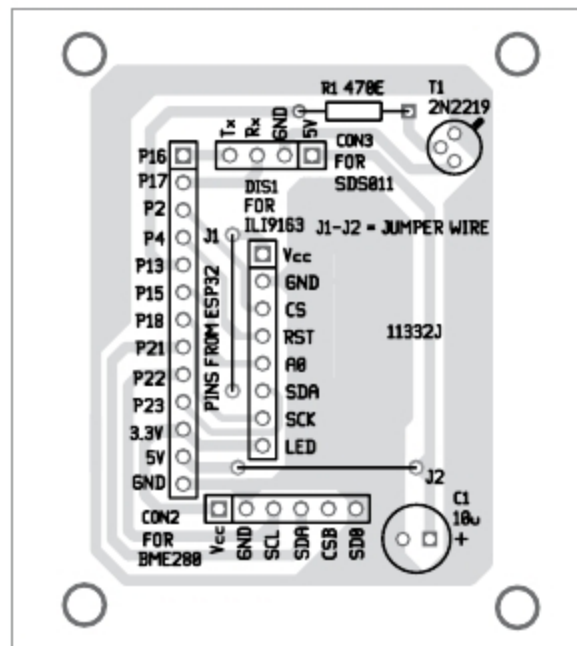


Fig. 8: Components layout for the PCB

Uploading the code to ESP32. Now everything is set and done by you. Assemble the devices on the PCB or general-purpose Veroboard. Connect the USB power supply cord to ESP32.

Run Arduino IDE and add the library for ESP32 and other relevant libraries. Open Arduino code (environment_thingspeak.ino).

Change/provide Wi-Fi ID and password towards the beginning of the sketch. Replace ThingSpeak channel number with your channel number, and API key with your API key.

Select ESP32 Dev Module from the pull-down menu under Tools→Board option. Next, select the correct USB COM port from the pull-down menu under Tools→Port option. Press Upload. If you have done everything right, the program will start communicating with ESP32 board and then upload the code into ESP32 MCU.

After a few seconds, you will see PM2.5, PM10 and other data on the TFT display screen. The same data

will also show up on the serial monitor of Arduino.

Open ThingSpeak channel on any Internet browser—your smartphone browser will work, too. You will find that data is stacking up perfectly, as shown in Fig. 6.

Note. You may refer to the following websites for details on instructions on how to integrate ESP32 on Arduino IDE:

<http://randomnerdtutorials.com/installing-the-esp32-board-in-arduino-ide-mac-and-linux-instructions/>

<http://randomnerdtutorials.com/installing-the-esp32-board-in-arduino-ide-windows-instructions/>

Construction and testing

An actual-size PCB layout for the air quality monitor is shown in Fig. 7 and its components layout in Fig. 8. Mount various modules on the PCB. Switch on the power supply to ESP32. If everything is correct, you will see the readings on the TFT display. The same values will appear on *www*.

EFY Note

The source code of this project is available for free download at source.efymag.com

PARTS LIST

Semiconductors:

T1 - 2N2219 npn switching transistor

Resistors (all 1/4-watt, ±5% carbon):

R1 - 470-ohm

Capacitors:

C1 - 10µF, 25V electrolytic

Miscellaneous:

CON1 - 8-pin connector for TFT display
CON2 - 6-pin connector for BME280
CON3 - 4-pin connector for SDS011
- ESP32 NodeMCU
- ILI9163, 3.6cm (1.44 inch) coloured TFT display
- SDS011 Nova dust sensor
- BME280 module

thingspeak.com, as shown in Fig. 6.

SDS011 runs on 5V. But, you can also use two 3.7V lithium-ion batteries along with a small linear regulator like 7805 to step-down to 5V. Alternately, you can use tiny converter (available on www.aliexpress.com) that can boost voltage from 1.5V to 5V. You can use one lithium-polymer battery along with a converter making it a truly portable device. **EFY**



Somnath Bera is an avid user of open source software. Professionally, he is a thermal power expert and works as additional general manager at NTPC Ltd

The author would like to dedicate this article to his mother, Late Mrs Sefali Bera, who passed away recently.

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